Cross-Corpus Speech Emotion Recognition Based on Dynamically Filtering Multistage Diffusion Model

Jingjie Yan, Boyan Sun, Yuebo Yue, Xiaoyang Zhou, and Ying Liu

***Abstract*—In cross-corpus speech emotion recognition (SER) tasks, although generative methods have shown certain potential in alleviating the scarcity of target domain samples, the generated samples often lack precise alignment with the acoustic-semantic features of the target domain. This results in pseudosamples of uneven quality and even introduces noise, thereby weakening the effectiveness of transfer learning. To address the problem of insufficient quality control over pseudosamples in existing methods, we propose a dynamically filtering multistage diffusion model (DFMDM). The proposed DFMDM model establishes a three-stage iterative optimization process of "generation–filtering–enhancement." In the first stage, the conditional diffusion model is guided by an adversarial domain classifier to generate pseudosamples with target domain characteristics. In the second stage, a dynamic filtering module is presented to eliminate low-quality pseudosamples that significantly deviate from the target domain features. In the third stage, the filtered pseudosamples are mixed with source domain samples and reused to regenerate target-style pseudosamples. In this way, a progressive diffusion-based enhancement loop is constructed, which improves sample quality through iterative generation and dynamic adjustment of similarity thresholds. Experimental results conducted on CASIA, EmoDB, eNTERFACE, and IEMOCAP demonstrate that the proposed DFMDM model achieves significant improvements over existing state-of-the-art (SOTA) approaches, fully showcasing its application potential and practical value in cross-corpus SER.**

***Impact Statement*—The current cross-corpus speech emotion recognition (SER) methods often lack accurate alignment of speech semantic features in the target domain, which weakens the performance of cross-corpus SER. By constructing the iterative optimization process of "generation–filtering–enhancement," this article proposes a dynamically filtering multistage diffusion model (DFMDM), which can make the generated speech samples closer to the target domain distribution and effectively improve the controllability and quality of pseudosamples. The experimental results show that the recognition rate of the DFMDM model is significantly higher than that of the existing SOTA method. The proposed model can be effectively applied to intelligent customer service, mental health, smart home and other fields in the field of artificial intelligence.**

***Index Terms*—Adversarial domain classifier, diffusion model, dynamic filtering module, speech emotion recognition (SER), transfer learning.**

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Jingjie Yan, Boyan Sun, Yuebo Yue, and Ying Liu are with the College of Telecommunications and Information Engineering, Nanjing University of Posts and Telecommunications, Nanjing 210003, China (e-mail: yanjingjie@njupt.edu.cn).

Xiaoyang Zhou is with the China Mobile Zijin (Jiangsu) Innovation Research Institute Company, Ltd., Southeast University, Nanjing 210096, China.

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I. Introduction

SPEECH emotion recognition (SER), as a focal area of intense research in the fields of affective computing and pattern recognition, seeks to empower computers to automatically analyze emotional states embedded in speech signals. Most existing methods are based on the premise that the training and testing datasets are identically and independently distributed, i.e., both follow similar statistical patterns. However, in real-world scenarios, factors such as variations in data acquisition equipment, linguistic and cultural diversity, heterogeneous distribution patterns, and imbalanced corpus sizes introduce significant domain shifts between the training and testing domains. These cross-domain discrepancies hinder traditional SER models from meeting the complex demands of real-world scenarios. Current research is mostly confined to single-corpus and monolingual settings, with insufficient systematic exploration of cross-domain generalization capabilities [1], [2]. Therefore, this article conducts a study on cross-corpus SER.

Current research on cross-corpus SER mainly focuses on techniques such as unsupervised domain adaptation [3], adversarial transfer learning [4], and generative data augmentation [5]. Unsupervised domain adaptation typically alleviates interdomain discrepancies through strategies such as feature alignment and distribution matching. However, these methods often rely on global statistics and neglect class-level alignment, which may lead to category confusion. Adversarial transfer approaches introduce domain discriminators to perform adversarial optimization between the source and target domains, thereby enhancing domain invariance of the model. Nonetheless, the training procedure frequently exhibits instability and prone to gradient vanishing problems. In generative augmentation methods, traditional adversarial-based generative models usually lack explicit control mechanisms, making it difficult to effectively guide the generated samples toward the target domain distribution. As a result, they struggle to generate representative samples with target-domain characteristics. In contrast, diffusion models [6] have risen to prominence as a highly effective category of generative techniques, offering stronger data modeling capacity and more stable sampling behavior. A diffusion probabilistic model comprises both a forward progression and a reverse procedure. The forward process gradually perturbs the original data by adding Gaussian noise until it becomes nearly isotropic, while the reverse process trains a neural network to iteratively remove noise and reconstruct the data manifold. This progressive denoising mechanism enables diffusion models to robustly capture complex structures and avoid mode collapse. By gradually guiding noise back toward the target data distribution, diffusion models can synthesize high-quality and structurally controllable emotional speech features, thereby improving the diversity and representativeness of generated samples and providing a more reliable augmentation strategy for cross-corpus recognition. To this end, the DACDM method [7] introduces a conditional diffusion model with a domain-guided mechanism that enables more precise control over the generation process, making the generated samples more closely aligned with the target-domain feature distribution. However, it still suffers from instability in the quality of generated samples, with intersample variance, which negatively impacts the overall performance and accuracy of subsequent cross-corpus emotion recognition tasks.

To address the aforementioned challenges, this article proposes a dynamically filtering multistage diffusion model (DFMDM), applied to cross-corpus SER. The model constructs an iterative optimization process of "generation–filtering–enhancement," comprising three stages in total. First, an adversarial training mechanism is integrated into the conditional diffusion model, where adversarial optimization between the source and target domains boosts the model's capacity to grasp the distinctive features of the target domain during the generation process. Second, a dynamic filtering module is proposed to assess the similarity of generated pseudosamples. Only high-quality samples that are highly consistent with target domain features are retained for subsequent training, thereby significantly reducing noise interference. Finally, the filtered samples are fused with source domain data to construct an enhanced source domain, which is then reinput into the diffusion model for the next round of generation. This enables a progressive improvement in both the quality and semantic consistency of pseudosamples.

Compared with existing approaches, the proposed method exhibits significant advantages in three aspects: 1) the introduction of an adversarial mechanism strengthens the domain-guidance capability of the diffusion model, making the generated samples more closely aligned with the target domain distribution; 2) the dynamic filtering strategy effectively improves the controllability and quality of pseudosamples, avoiding the negative impact of low-quality samples on recognition performance; and 3) the iterative generative mechanism of "generation–filtering–enhancement" gradually narrows the distribution gap between generated samples and the target domain, thereby enhancing the realism and task adaptability of the samples.

The main contributions of this article are summarized.

1) A dynamic filtering module is proposed to evaluate the similarity of generated pseudosamples. Only high-quality samples that are highly consistent with target domain features are retained for subsequent training, which significantly reduces noise interference.
2) A DFMDM is introduced. By constructing a three-stage iterative optimization process, a progressive diffusion-based enhancement loop is formed during pseudosample generation, where the similarity threshold is dynamically adjusted to improve the quality of generated samples.
3) Extensive cross-corpus experiments are conducted on the eNTERFACE, CASIA, EMO-DB, and IEMOCAP databases to validate the effectiveness of the proposed method.

II. Related Work

A. Research on SER

In SER tasks, various machine learning-based classifiers play a critical role. Representative methods include support vector machines (SVM) [8], Gaussian mixture models (GMM) [9], and artificial neural networks (ANN) [9]. Despite the respective advantages of these approaches, they still face limitations in addressing issues such as data imbalance and cross-lingual adaptability [10].

In recent times, deep learning approaches have made remarkable strides in the domain of SER, due to the powerful capabilities in multilevel feature representation. The main models include convolutional neural networks (CNN) [11], long short-term memory networks (LSTM) [12], and Transformer networks [13], [14]. Among these, CNNs effectively extract local spatial structure information from speech spectrograms, demonstrating strong noise robustness and cross-lingual adaptability. LSTMs capture long-term contextual dependencies in speech sequences, exhibiting outstanding performance particularly in conversational scenarios. Li et al. [13] employ Transformer models to further extract emotion-related features directly from raw speech signals. Akinpelu et al. [14] transform speech signals into spectrograms and then feed them into vision transformers (ViT) to extract speech emotion features from the spectral domain. However, current deep learning models still face challenges such as high computational resource consumption, unstable performance in natural environments, and lack of interpretability [10].

B. Research on Cross-Corpus SER

Deep learning-based methods typically rely on large-scale data and massive computational resources, whereas transfer learning can leverage pretrained models from existing databases to rapidly build machine learning models for new datasets. In recent years, cross-corpus SER has garnered considerable interest from both academic circles and the industrial sector as a key branch of transfer learning. DIDAN [15] constructs

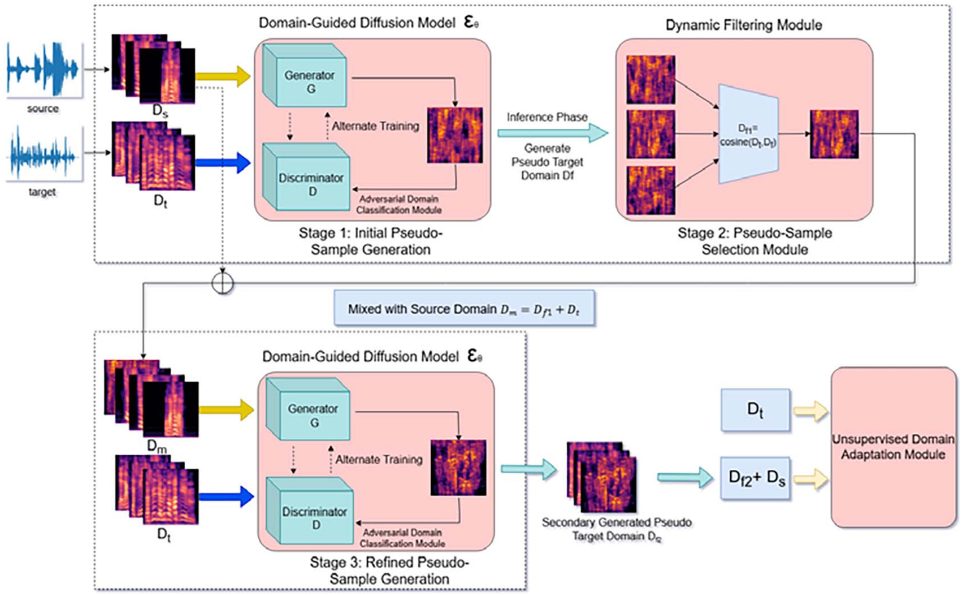

Fig. 1. Network diagram of DFMDM.

a deep regression network that incorporates convolutional and fully-connected layers, initially acquiring emotional discriminative ability by mapping source speech spectrograms to emotion labels. Cai et al. [4] constructed a cross-domain emotion recognition framework leveraging an adaptive adversarial transfer network. This model dynamically assesses the relative contributions of global domain distribution and local semantic distribution via an adaptive adversarial module. Li et al. [16] proposed a multisource discriminative subspace alignment (MDSA) method. MDSA constructs discriminative subspaces for multiple source domains and dynamically optimizes the target subspace representation via an adaptive weight-driven linear reconstruction mechanism. Gao et al. [17] introduced the adversarial domain generalization transformer (ADoGT), which eliminates domain-specific interference through an adversarial feature disentanglement mechanism to construct a cross-domain emotional feature space. Tang et al. [5] proposed a hierarchical emotional speech synthesis framework ED-TTS, achieving fine-grained control of speech emotional expression through multigranularity modeling. The method innovatively constructs a dual-path emotional representation system. Zhao et al. [3] proposed an emotion-aware contrastive adaptation network (ECAN) for unsupervised cross-corpus SER, focusing on the cooperative optimization of local sample structures and global class distributions during domain adaptation. Zhao et al. [18] studied an acoustic knowledge transfer learning with regularization (AKTLR) framework, focusing on the collaborative optimization of acoustic features and transfer learning for cross-corpus SER.

III. DFMDM

A. Network Overview

As illustrated in Fig. 1, this article proposes a DFMDM to improve the generation quality and semantic consistency of pseudotarget domain samples for cross-corpus SER. The overall framework follows a three-stage pipeline: 1) generation; 2) filtering; and 3) enhancement.

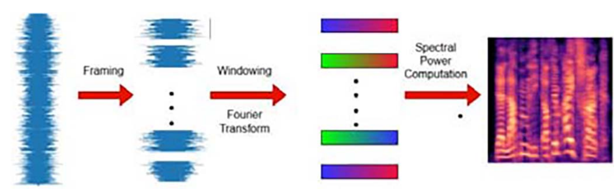

Fig. 2. Spectrogram generation process.

Stage I (Generation) focuses on generating initial pseudotarget domain samples with target-domain characteristics. Specifically, speech signals from the source domain D_s and the target domain D_t are first preprocessed into spectrograms, which serve as inputs to the conditional diffusion model. During the training phase, an adversarial domain discriminator is introduced to perform adversarial optimization between the source and target domains, guiding the diffusion model to generate pseudosamples D_f that capture the distinctive distributional features of the target domain. During the inference phase, the domain discriminator is kept frozen and incorporated into the diffusion model to guide the pseudosample generation process through a domain-guided strategy.

Stage II (Filtering) aims to improve the quality and reliability of the generated pseudosamples. A CNN is employed to extract deep features from both the pseudotarget domain samples D_f and the real target domain samples D_t. The cosine similarity between these features is then computed, and low-quality pseudosamples with low confidence are filtered out. The remaining high-confidence pseudosamples are retained and fused with the original source domain samples D_s to construct a mixed source domain dataset D_m. This stage is applied only during training and is excluded from the inference process.

Stage III (Enhancement) further enhances the semantic and distributional alignment between the pseudosamples and the target domain. During training, the mixed source domain D_m and the target domain D_t are jointly fed into the conditional diffusion model for a second round of generation, producing refined pseudotarget domain samples D_{f2}. Finally, the high-quality pseudotarget domain samples D_{f2}, optimized through two rounds of generation and filtering, are combined with the source domain samples for model training, significantly improving the generalization ability and robustness of cross-domain recognition.

B. Speech Signal Feature Extraction

To facilitate subsequent processing and modeling of speech signals, the original audio data is first converted into spectrograms [19]. Fig. 2 illustrates the spectrogram generation process. The speech signal is first segmented into fixed-length short-time frames through framing to capture its local stationarity. Then, a window function (e.g., Hamming window) is applied to mitigate spectral distortion arising from truncation effects. Each framed time-domain signal undergoes transformation into the frequency domain via FFT, and the squared magnitude of the spectrum is computed to obtain the energy distribution, which is further converted to a decibel scale to enhance visualization contrast. Finally, the power spectra of all frames are arranged sequentially in time to form the two-dimensional time-frequency spectrogram. In this work, the audio is sampled at 16 kHz, and the spectrogram is computed with a window duration of 25 ms, a frame advancement of 10 ms, a 512-point FFT, and 64 Mel filter banks.

C. Domain-Guided Diffusion Model

In unsupervised domain adaptation tasks, DACDM [7] achieves dual controllability over the semantic category and domain distribution features of generated samples through the collaborative design of a conditional diffusion mechanism and a domain guidance strategy. The conditional diffusion mechanism introduces adaptive group normalization layers (AdaGN) into the UNet architecture of the diffusion model $\hat{\epsilon}_\theta(x_t, t)$, encoding class label information (either ground-truth source domain labels or pseudolabels from the target domain) as conditional embeddings injected into the intermediate network layers. This enables the model to learn semantic features of the data during the denoising process. For source domain samples, the generation process is directly constrained by the true labels to ensure semantic accuracy. For unlabeled target domain samples, pseudolabels generated by a pretrained unsupervised domain adaptation (UDA) model are used as conditional inputs to compensate for the lack of target domain labels. The training objective for pseudolabels in the conditional diffusion model is to minimize [7]

$$L_{t\sim\mu(1,T),\, x_0\sim D_s\cup D_t,\, \varepsilon\sim\mathcal{N}(0,I)}\ \Omega_t \left\|\varepsilon - \varepsilon_\theta(x_t, t, c)\right\|^2 \tag{1}$$

where x_0 denotes a sample drawn from either the source domain or the target domain, $\varepsilon \sim \mathcal{N}(0, I)$ represents Gaussian noise, x_t is the noisy sample at the tth step, c is the class condition (either a ground-truth label or a pseudolabel), and $\varepsilon_\theta(x_t, t, c)$ denotes the model's prediction of the noise.

The domain guidance strategy trains a domain classifier ϕ to distinguish feature distributions of source and target domains on the intermediate diffusion state x_t, and uses its gradient signal to correct the reverse diffusion sampling process. During generation, a domain guidance term is introduced by integrating the gradient information from the domain classifier into noise prediction, forcing the domain probability of generated samples to converge to the target domain label d_t

$$\begin{aligned}\hat{\epsilon}_\theta(x_t, t, \hat{c}) &= \epsilon_\theta(x_t, t, y)\\ &\quad - \sigma\sqrt{1-\bar{\alpha}_t}\nabla_{x_t}\log p_{\phi^*}(d_t|x_t, t)\end{aligned} \tag{2}$$

where $\hat{c} = [y, d_t]$ is the joint encoding of the class label y and target domain label d_t, σ is the guidance strength coefficient, $\bar{\alpha}_t$ represents the time-step-wise cumulative product of α_t, representing the total signal preserved up to step t, and $p_{\phi^*}(d_t|x_t, t)$ is the domain classifier's predicted probability of the target domain. Through gradient backpropagation, the domain features of generated samples are forced to approach the target domain, achieving directional alignment of domain distributions.

D. Adversarial Domain Classification Module

In conditional diffusion models, although the emotional category of generated samples can be controlled via class labels [20], it cannot be guaranteed that the generated samples possess the characteristic attributes of the target domain. Therefore, a domain discriminator module is introduced to provide domain-level guidance during the diffusion process, facilitating the creation of samples that closely match the target domain distribution. However, the domain classifiers adopted in methods such as DACDM [7] typically operate in a static manner, meaning that the classifier is pretrained before the generation stage and its parameters remain fixed throughout the diffusion process. This static nature leads to several limitations. On the one hand, a pretrained, fixed classifier cannot adapt to the evolving distribution of generated samples, resulting in weak coupling with the diffusion generation process and a lack of continuous feedback, which makes effective regulation of the generation trajectory difficult. On the other hand, when there are significant distribution discrepancies or class imbalances between the source and target domains, the classifier tends to be biased toward the source domain, reducing the domain consistency of the generated samples.

To address this, this article introduces an adversarial training-based domain classification mechanism within the conditional diffusion framework, replacing the traditional static discriminator. This approach draws inspiration from generative adversarial networks (GAN) [21], [22] and domain-adversarial neural networks (DANN) [21], establishing an adversarial optimization relationship between the source and target domains to dynamically guide the generation process, and enhance the diffusion model's ability to capture target domain features and guide the generation of more adaptable pseudosamples.

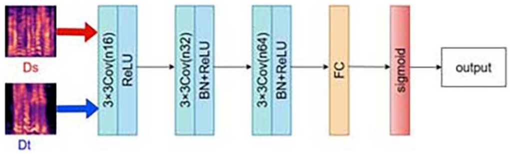

Fig. 3. Discriminator network model diagram.

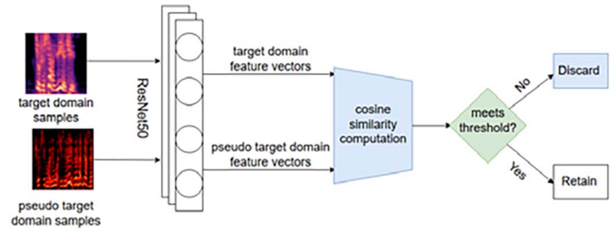

Fig. 4. Dynamic threshold filtering network architecture.

In this approach, the constructed discriminator is used to differentiate whether an input sample originates from the source domain D_s or the target domain D_t. The discriminator is structured as a multilayer perceptron (MLP) model, producing a scalar output $F(x) \in [0, 1]$. When $F(x)$ approaches 1, the input is identified as belonging to the target domain. When it approaches 0, it is classified as the source domain. The training objective of the discriminator is to maximize classification accuracy between source and target domain samples, with the optimization objective defined as

$$\max_{D} V(D_s, D_t) = \log F(D_s) + \log(1 - F(D_t)) \tag{3}$$

where the first term $\log F(D_s)$ is the log-probability of correctly classifying source domain samples, and the second term $\log(1 - F(D_t))$ is the log-probability of correctly identifying target domain samples.

The inputs, consisting of samples from both the source and target domains, first undergo a 3×3 convolutional layer featuring n_{16} channels, followed by a ReLU activation function. Next, they enter a second 3×3 convolutional layer featuring n_{32} channels, which is succeeded by batch normalization and a subsequent ReLU activation to stabilize training and activate features. Then, the data proceeds through a third 3×3 convolutional layer featuring n_{64} channels, also succeeded by batch normalization and a subsequent ReLU activation. Finally, the features are integrated through a FC layer succeeded by a sigmoid activation function, which maps the output to a scalar value representing the domain distribution probability, ranging between 0 and 1. The diagram of discriminator network model is illustrated in Fig. 3. Once the discriminator has been trained to effectively differentiate between samples from the source and target domains, it is incorporated into the diffusion model as the domain classifier ϕ. By embedding ϕ into the denoising network $\hat{\epsilon}_\theta$, domain-level feedback can be provided during the generation process, enabling the diffusion model to synthesize pseudosamples that better align with the target domain.

E. Dynamic Filtering Module

Owing to the intrinsic stochasticity present in the diffusion-based generation procedure, the quality of pseudotarget domain samples generated in the first stage varies and may include semantically shifted or unrepresentative samples. Directly using all pseudosamples could introduce noise and adversely affect the subsequent model's transfer performance. Therefore, it is necessary to screen the generated samples and retain only high-quality pseudosamples that are highly consistent with the true target domain features. This study proposes a dynamic filtering module based on feature similarity to evaluate the semantic similarity between pseudosamples and real target domain samples. The specific implementation steps are as follows.

First, a lightweight CNN is employed as a feature extractor to obtain spectrogram feature representations of each generated sample and target domain sample. Then, cosine similarity [23] is used as the similarity metric to calculate the feature similarity between generated and target domain samples. For each generated sample, if its average similarity to the target domain samples exceeds a preset threshold, it is regarded as a high-quality sample and retained. Otherwise, it is discarded. The cosine similarity is

$$\cos(\theta) = \frac{\sum_{j=1}^{n} A_j \times B_j}{\sqrt{\sum_{j=1}^{n} A_j^2} \times \sqrt{\sum_{j=1}^{n} B_j^2}} \tag{4}$$

where the vector representations of the generated sample and target domain sample in the feature space are $\vec{A} = (A_1, A_2, \ldots, A_n)$ and $\vec{B} = (B_1, B_2, \ldots, B_n)$, respectively. The numerator $\sum_{j=1}^{n} A_j \times B_j$ represents the dot product of the two vectors, characterizing their coordinated variation across dimensions, the denominator $\sqrt{\sum_{j=1}^{n} A_j^2} \times \sqrt{\sum_{j=1}^{n} B_j^2}$ are the L2 norms of vectors $\vec{A}$ and $\vec{B}$, representing their magnitudes. The cosine similarity spans from 0 to 1, where higher values signify a stronger degree of similarity. Through this screening mechanism, the system dynamically controls the quality and quantity of pseudosamples, ensuring semantic consistency while maintaining diversity of generated samples, thereby establishing a robust groundwork for the subsequent enhanced generation and transfer learning stages. The diagram of dynamic filtering module is illustrated in Fig. 4.

F. Pseudotarget Domain-Driven Third-Stage Conditional Diffusion Generation Mechanism

After completing the first-stage diffusion generation and similarity-based screening, the system obtains a set of pseudotarget domain samples exhibiting excellent semantic representation and domain characteristics. Although these samples possess a certain degree of representativeness, due to the inherent stochastic perturbations in the diffusion process, there may still be issues such as fine-grained information loss and local structural shifts, resulting in insufficient fitting capability to the target domain distribution. To address this, this article further proposes a third-stage diffusion generation mechanism guided by the pseudotarget domain as a condition, aiming to

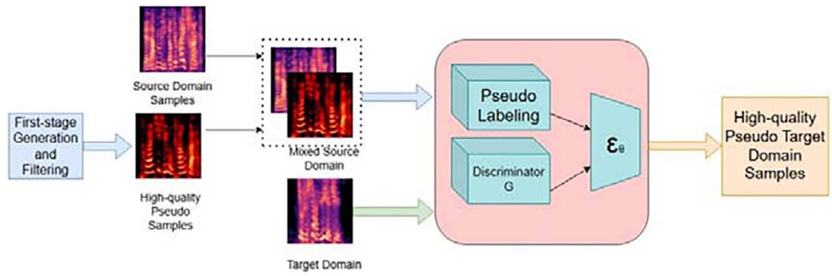

Fig. 5. Pseudotarget domain-driven third-stage conditional diffusion generation diagram.

achieve dual optimization of generation quality and distribution consistency.

The core idea of this mechanism is to fuse the high-quality pseudosamples generated in the first stage and retained after screening in the second stage with the real source domain samples, forming a new enhanced mixed source domain dataset, which serves as the conditional input for the diffusion model in the third-stage generation. By introducing training samples with a stronger target domain style, the diffusion model can learn representations closer to the target domain during retraining, thereby enabling the newly generated samples to further improve their domain adaptability while maintaining emotional semantic consistency.

During the diffusion modeling process, category labels and domain guidance are still used as conditional inputs. However, unlike the first stage, the guidance conditions in the third stage exhibit stronger target domain transfer properties. On one hand, they originate from the screened pseudosamples. On the other hand, they incorporate source domain label information. This semisupervised hybrid conditioning guides the diffusion model to gradually migrate toward the target domain features in the latent space, reinforcing semantic consistency and distribution alignment of the generated samples. Through the "generation–screening–enhancement" mechanism, the distribution gap between generated samples and the target domain is progressively narrowed across iterations, forming an error-reducing optimization trajectory. The model deeply integrates the high-fidelity sample generation capability of conditional diffusion, the similarity threshold filtering mechanism, and iterative enhancement with dynamic feedback, constructing a progressive optimization process that addresses complex domain discrepancies in cross-corpus SER. This approach preserves the annotation advantages of the source domain while exponentially improving the quality of generated samples via target domain distribution-guided diffusion, thus providing high-quality data with both rich annotations and domain adaptability for subsequent transfer learning.

Ultimately, the pseudotarget domain samples produced during the third phase not only exhibit a more pronounced target domain style visually but also demonstrate higher stability in discriminating emotional categories. This module, as a pseudosupervised conditional regeneration mechanism, forms a critical component of the progressive cross-domain optimization strategy proposed in this article. The diagram of pseudotarget domain-driven third-stage conditional diffusion generation mechanism is illustrated in Fig. 5.

TABLE I
SAMPLE DISTRIBUTION (ORI IS ORIGINAL SAMPLES AND GEN IS GENERATED SAMPLES)

Emotion Category	CASIA		EmoDB		eNTERFACE	
	Ori.	Gen.	Ori.	Gen.	Ori.	Gen.
Angry	200	200	127	120	215	215
Fear	200	200	69	60	215	215
Happy	200	200	71	70	215	215
Sad	200	200	62	60	215	215
Total	800	800	329	310	860	860

G. Unsupervised Domain Adaptation Module

After completing diffusion generation, similarity screening, and the secondary diffusion generation, the final generated pseudotarget domain samples are merged alongside the samples from the source domain to constitute an enhanced mixed source domain D_m. This mixed source domain along with the target domain D_t are inputted into the unsupervised domain adaptation module to perform cross-corpus SER. This study adopts the multirepresentation adaptation network (MRAN) [24], whose multirepresentation feature extraction module extracts features covering multiple dimensions such as texture, color, and shape through multiple substructures, overcoming the information limitations of single structures. Meanwhile, the maximum mean discrepancy (MMD) is extended to conditional maximum mean discrepancy (CMMD) [24], which aligns class-conditional distributions by utilizing source domain labels and target domain pseudolabels, enforcing intraclass consistency in the subspace after domain adaptation and addressing the shortcomings of marginal distribution alignment. The diagram of unsupervised domain adaptation with MRAN [24] is illustrated in Fig. 6.

Finally, the detailed process and steps of the proposed DFMDM model is summarized in Algorithm 1.

IV. CROSS-DATABASE EXPERIMENTS

This section introduces the speech emotion databases used in the experiments, which include CASIA [25], EmoDB [26], eNTERFACE [27], and IEMOCAP [28]. The sample distribution for each database can be referred to in Table I.

A. Databases

1) eNTERFACE: The eNTERFACE database [27], is a bimodal emotion database integrating both speech and video data, aimed at providing a standardized resource for emotion recognition research. The research team collects data from 46 subjects

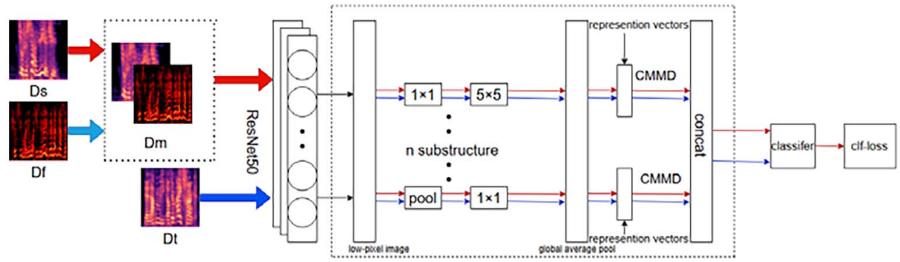

Fig. 6. Unsupervised domain adaptation with MRAN diagram [24].

from 14 countries (81% male, 19% female) in a standardized experimental environment. Each subject is instructed to naturally express six emotions—1) happy; 2) sad; 3) surprise; 4) angry; 5) disgust; and 6) fear—in six different scenarios. After data collection, human experts review each sample; remove those with ambiguous emotional expressions, resulting in valid data from 42 subjects.

2) EmoDB: EmoDB [26] focuses on the construction and characteristics of a German emotional speech database. It encompasses audio recordings sourced from 10 actors (five male and five female performers), mimicking seven target emotions: 1) neutral; 2) angry; 3) fear; 4) happy; 5) sad; 6) disgust; and 7) boredom. The database contains approximately 800 emotional speech samples, recorded from 10 daily communication sentences (five short and five long).

3) CASIA: CASIA [25] is recorded in a professional studio environment by female professional speakers. The dataset contains 5000 sentences balanced in speech context, totaling approximately 7 h. Its textual annotations include word boundaries, part-of-speech tags, and pronunciation information. It covers happy, sad, surprise, angry, and disgust category.

4) IEMOCAP: IEMOCAP [28] stands as a commonly utilized multimodal and speech emotion database. In this study, to ensure consistency with previous work [29], only four emotion categories, namely angry, happy, neutral, and sad, are considered. The excited category is merged into the happy class. After filtering, the resulting subset contains a total of 5531 speech utterances, including 1103 angry emotion samples, 1636 happy emotion samples, 1084 sad emotion samples, and 1708 neutral emotion samples.

Since the emotion categories across different databases are not fully aligned, to ensure the compatibility and validity of cross-database experiments as well as fairness in comparison with baseline methods, two experimental strategies are adopted in this study. First, experiments are conducted using all shared emotion categories between the source and target domains, and the corresponding results are reported in Table II(A). Second, four fundamental emotions that are common to all databases—angry, fear, happy, and sad—are selected as unified target categories. The results under this setting are presented in Table II(B), based on which further ablation studies are conducted.

B. Experimental Design

The hardware equipment of the experiment is mainly based on NVIDIA RTX 4090d GPU. The experiments use the Adam optimization algorithm with a batch size configured to 32, an initial learning rate of 0.0001, a weight decay of 1e-8, and a Dropout probability of 0.5. The model undergoes training for a total of 200 epochs.

The cross-database SER experiments include eight transfer tasks: 1) C→B; 2) C→E; 3) B→C; 4) B→E; 5) B→I; 6) I→B; 7) E→C; and 8) E→B, where C, B, I, and E denote CASIA, EmoDB, IEMOCAP, and eNTERFACE, respectively. Following standard cross-domain evaluation protocols, UAR and WAR [30] are adopted as the main metrics. To clarify the experimental configuration, we take the B→C task as an example. In this setting, EmoDB serves as the source domain and CASIA as the target domain. During the diffusion-model training stage, source-domain samples are used with their ground-truth labels, while target-domain samples, originally unlabeled, are assigned pseudolabels by a pretrained UDA model and are jointly utilized to guide the diffusion model toward the target emotional distribution. After this stage, the generated high-quality pseudotarget samples and the original source data are merged to form the mixed labeled set. This mixed set—together with the unlabeled target-domain samples—is subsequently used as the input for the domain-adaptation training stage, where the model further aligns the two domains in a fully unsupervised manner. For evaluation, all CASIA samples with true labels serve as the testing set. The same experimental protocol is applied to all other transfer tasks with the corresponding source–target assignments.

C. Experimental Results and Analysis

As shown in the table, the proposed DFMDM model exhibits consistently strong performance across diverse cross-lingual transfer settings. Representative source-free approaches such as ECAN and USFAN rely exclusively on target-domain feature adaptation without accessing source-domain data. However, the absence of source-domain information restricts their ability to discriminate domain-specific characteristics, resulting in limited robustness when facing substantial tonal and prosodic disparities across languages. In contrast, the proposed method integrates conditional diffusion-based high-fidelity sample generation with a progressive diffusion-enhanced iterative loop, and further incorporates cosine-threshold dynamic similarity selection and MRAN-driven multirepresentational feature extraction. These components jointly strengthen cross-lingual feature alignment and effectively enhance emotion classification performance under challenging cross-domain and cross-language scenarios.

Algorithm 1: DFMDM Model.

Input: Source domain D_s, target domain D_t, maximum iteration $N_{\max}$, similarity threshold τ
Output: Trained conditional diffusion model $\hat{\varepsilon}_\theta$ and refined pseudo target samples D_{f2}
Initialization:
Preprocess speech signals from D_s and D_t into spectrogram representations;
Initialize UDA model parameters θ_f, diffusion model parameters θ, and domain classifier parameters ϕ;
Set iteration index $n \leftarrow 1$;
Step 1: UDA Training
Train UDA model $f^{(n)}(x)$ using labeled D_s and unlabeled D_t;
Step 2: Pseudo-label Generation
Generate pseudo-labels $\hat{y}_t^j = f^{(n)}(x_t^j)$ for each $x_t^j \in D_t$;
Construct mixed dataset $D_{\text{mix}} = D_s \cup \{(x_t^j, \hat{y}_t^j)\}$;
while $n <= N_{\max}$ **do**
Step 3: Conditional Diffusion Training
Train conditional diffusion model $\varepsilon_\theta(x_t, t, c)$ on D_{mix} by minimizing:

$$L_{t\sim\mu(1,T),\, x_0\sim D_s\cup D_t,\, \varepsilon\sim\mathcal{N}(0,I)}\ \Omega_t\|\varepsilon - \varepsilon_\theta(x_t,t,c)\|^2$$

until diffusion loss converges or maximum diffusion steps are reached;
Step 4: Domain Adversarial Guidance
Train domain classifier $\phi^{(n)}$ by maximizing:

$$\max_D V(D_s, D_t) = \log F(D_s) + \log(1 - F(D_t))$$

until adversarial equilibrium is achieved;
Step 5: Pseudo Sample Generation and Filtering
Generate target-domain pseudo samples D_f using adversarially guided diffusion:

$$\hat{\varepsilon}_\theta = \varepsilon_\theta - \sigma\sqrt{1-\bar{\alpha}_t}\nabla_{x_t}\log p_{\phi^{(n)}}(d_t|x_t,t)$$

Compute cosine similarity $\cos(D_f, D_t)$ and retain samples $D_f n$ satisfying $\cos(\cdot) > \tau$;
Form augmented mixed-domain dataset $D_{\text{mix}} = D_{\text{mix}} \cup D_f n$;
$n \leftarrow n+1$;
Step 6: Final Adaptation:
Construct global mixed source domain $D_g = D_s \cup D_{fN}$;
Perform final UDA training on D_g and D_t until validation loss converges or maximum epochs are reached;
return $\hat{\varepsilon}_\theta$, D_{fN};

Further analysis of experimental data shows that in the target domain sample-scarce E→B scenario, DFMDM achieves a UAR of 53.99%, surpassing USFAN (51.73%) and NRC (48.42%), thereby validating the effectiveness of diffusion models in generating high-quality pseudosamples by simulating the target domain distribution and alleviating overfitting caused by small samples. Moreover, the experimental results in Table III reveal that DFMDM remains highly competitive when transferring between large-scale and small-scale datasets. In summary, through a generate–select–enhance closed loop, DFMDM overcomes the limitations of traditional adversarial and subspace approaches. Its performance advantage is particularly notable under complex domain discrepancies and low-resource conditions.

Moreover, the confusion matrices of DFMDM are presented in Fig. 7. The model achieves consistently stronger performance on EmoDB, largely due to the pseudotarget samples synthesized by DFMDM, which help alleviate overfitting in small-scale corpora and support more stable learning of emotional features. Moreover, cross several cross-corpus settings, noticeable confusion occurs between the *fear* and *sad* categories. As these emotions share similar negative valence and partially overlapping acoustic cues, distribution shifts further blur their boundaries, leading to higher misclassification rates.

To further demonstrate the robustness and generalization capability of the proposed method, this study uses WAR as a supplementary metric for three SOTA methods: L-DA [39], MDSA [40], and DADR [41]. The experimental results of WAR (in Table IV) demonstrate that the proposed method achieves the best performance across all tasks. For example, in the C→B

TABLE II
COMPARATIVE EXPERIMENTAL RESULTS (UAR %)

Model	B→C	C→B	B→E	E→B	C→E	E→C
(A) Using All Shared Classes Between Source and Target Domains						
DAN [22]	36.30	56.72	33.58	43.50	32.17	29.30
SHOT [31]	34.80	55.83	31.92	46.08	30.99	31.80
G-SFDA [32]	27.90	50.41	23.21	35.79	27.09	24.30
NRC [33]	37.80	60.16	31.87	48.42	32.22	31.40
USFAN [34]	32.70	53.72	30.77	51.73	30.60	29.00
CoWA-JMDS [35]	34.50	55.26	28.49	36.74	31.61	25.80
DaC [36]	31.90	47.42	26.42	39.03	25.60	27.00
AaD [37]	35.00	55.50	31.47	48.12	32.17	29.10
LIDAN [38]	39.00	58.06	34.44	47.04	**34.97**	33.88
AKTLR [18]	**45.00**	59.93	32.51	43.60	34.09	37.60
ECAN [3]	39.00	61.37	34.21	46.87	34.53	31.90
Ours	42.50	**63.33**	**34.78**	**53.99**	33.48	**38.86**
(B) Using Only the Common Classes Across All Three Databases						
L-DA [39]	38.16	58.13	31.77	44.61	34.29	30.11
MDSA [40]	41.44	65.59	34.95	49.02	35.77	32.43
DADR [41]	40.53	63.61	36.93	47.38	38.29	34.98
DACDM [7]	41.03	62.17	35.41	47.87	37.35	33.92
Ours	**44.62**	**66.66**	**37.21**	**51.95**	**40.00**	**36.13**

Note: Bold entries represent the best experimental results.

TABLE III
EXPERIMENTAL RESULTS BETWEEN IEMOCAP AND EMO-DB (WAR %)

Model	**B→I**	**I→B**
GSDA-PCA [42]	44.57	66.75
UTCL [43]	47.41	66.13
CDLSL [44]	46.81	67.35
L-DA [39]	47.51	39.53
MDERE [29]	50.26	63.93
DACDM [7]	46.52	65.49
Ours	**51.38**	**69.03**

Note: Bold entries represent the best experimental results.

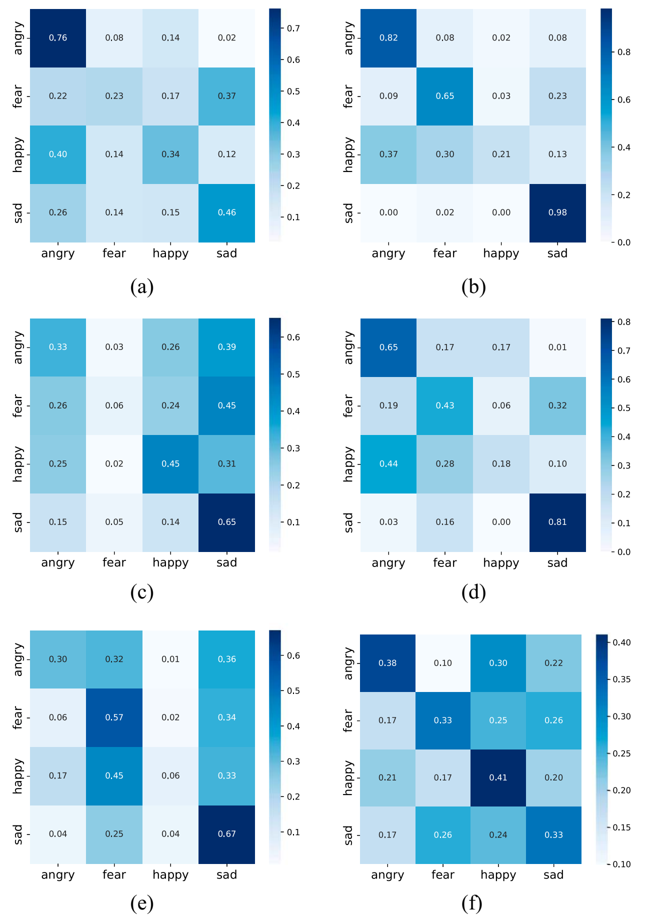

Fig. 7. Confusion matrices of the proposed DFMDM model in the six cross-database SER tasks. (a) B-C; (b) C-B; (c) B-E; (d) E-B; (e) C-E; and (f) E-C.

TABLE IV
COMPARATIVE EXPERIMENTAL RESULTS (WAR %)

Model	B→C	C→B	B→E	E→B	C→E	E→C
L-DA [39]	38.16	60.16	31.77	45.13	34.29	30.11
MDSA [40]	41.44	65.59	34.95	49.02	35.77	32.43
DADR [41]	40.53	63.61	36.93	47.38	38.29	34.98
Ours	**44.62**	**68.39**	**37.21**	**53.50**	**40.00**	**36.13**

Note: Bold entries represent the best experimental results.

task, DFMDM attains a UAR of 66.66%, showing significant improvement compared with DADR (63.61%), MDSA (65.59%), and L-DA (58.13%). Meanwhile, the WAR reaches 68.39%, which is substantially higher than other methods. Moreover, in balanced database transfer tasks such as B→E and C→E, the proposed method also exhibits a clear advantage in WAR. CASIA and eNTERFACE databases have relatively balanced class distributions, whereas EmoDB is a typical imbalanced dataset. Despite this, the proposed DFMDM still significantly outperforms others in WAR for imbalanced scenarios (e.g., C→B and E→B tasks).

D. Ablation Study

In the proposed DFMDM model, dynamic filtering is applied to the generated pseudotarget-domain samples, where cosine similarity is adopted as the filtering criterion. This design enables ablation studies by adjusting different similarity thresholds to analyze their impacts on the performance of the model. To investigate the effectiveness and sensitivity of the dynamic filtering module, a series of experiments are conducted with different cosine similarity thresholds (0.3, 0.4, 0.5, 0.6, and 0.7), and both UAR and WAR are utilized as assessment criteria. The experimental results are reported in Tables V and VI. The results indicate that the optimal similarity threshold varies across different cross-database transfer scenarios due to the inherent differences in data distributions, recording conditions, and emotional expression patterns among databases. From the perspective of the UAR evaluation results, a similar trend can be consistently observed across different cross-database transfer tasks. Specifically, the optimal threshold is 0.5 for the B→C and E→C tasks, 0.4 for the C→B task, and 0.6 for the B→E, E→B, and C→E tasks. Overall, the experimental results suggest that a cosine similarity threshold in the range of 0.4–0.6 provides a favorable balance between pseudosample quality and quantity, leading to consistently better recognition performance across most cross-database tasks. In contrast, a lower threshold (e.g., 0.3) fails to effectively filter out noisy pseudosamples, resulting in degraded sample quality and inferior model performance. Conversely, an excessively high threshold (e.g., 0.7) overly restricts the number of retained pseudosamples and leads to noticeable performance degradation in certain transfer tasks (e.g., E→C).

TABLE V
UAR % RESULTS UNDER DIFFERENT THRESHOLDS

Threshold	B→C	C→B	B→E	E→B	C→E	E→C
0.3	35.26	54.40	31.86	43.97	34.64	28.25
0.4	40.75	**66.66**	33.02	48.38	36.51	35.63
0.5	**44.62**	65.01	36.98	50.04	38.26	**36.13**
0.6	44.37	63.61	**37.21**	**51.95**	**40.00**	35.88
0.7	42.63	57.84	35.47	49.95	38.49	28.21

Note: Bold entries represent the best experimental results.

TABLE VI
WAR % RESULTS UNDER DIFFERENT THRESHOLDS

Threshold	B→C	C→B	B→E	E→B	C→E	E→C
0.3	35.26	56.84	31.86	50.46	34.65	28.25
0.4	40.75	66.66	33.02	48.38	36.51	35.63
0.5	**44.62**	**68.39**	36.98	51.67	38.26	**36.13**
0.6	44.37	63.61	**37.21**	**53.50**	**40.00**	35.88
0.7	42.63	57.84	35.47	51.23	38.49	28.21

Note: Bold entries represent the best experimental results.

These observations demonstrate that the proposed dynamic filtering module effectively balances sample quality and quantity, thereby significantly enhancing the robustness and generalization capability of cross-database SER. The above analysis further validates the effectiveness and necessity of the similarity-based selection mechanism in the proposed DFMDM framework. To thoroughly assess the impact of every crucial element in the proposed method, four groups of ablation experiments are further designed from the perspectives of the domain guidance strategy, pseudosample filtering mechanism, regeneration process, and overall training workflow.

TABLE VII
UAR % RESULTS OF DIFFERENT STRATEGIES

Strategy	B→C	C→B	B→E	E→B	C→E	E→C
w/o AG	28.25	48.17	25.33	38.56	29.63	22.14
w/o SF	43.34	65.18	36.68	49.78	38.51	35.63
w/o SG	42.16	63.32	35.98	48.59	38.12	34.83
DMF	43.72	64.87	36.13	50.32	38.43	35.26
Ours	**44.62**	**66.66**	**37.21**	**51.95**	**40.00**	**36.13**

Note: Bold entries represent the best experimental results.

TABLE VIII
WAR % RESULTS OF DIFFERENT STRATEGIES

Strategy	B→C	C→B	B→E	E→B	C→E	E→C
w/o AG	28.25	49.65	25.33	39.14	29.63	22.14
w/o SF	43.34	66.27	36.68	51.23	38.51	35.63
w/o SG	42.16	63.96	35.98	50.14	38.12	34.83
DMF	43.72	65.49	36.13	51.86	38.43	35.26
Ours	**44.62**	**68.39**	**37.21**	**53.95**	**40.00**	**36.13**

Note: Bold entries represent the best experimental results.

The first group, *w/o adversarial guidance* (w/o AG), removes the adversarial domain classifier in the diffusion generation process, retaining only the emotion category as the generation condition, to evaluate the effectiveness of the adversarial mechanism in guiding the generation of target domain features. The second group, *w/o similarity filtering* (w/o SF), skips the second-stage sample filtering after the first-stage diffusion generation and directly uses all pseudosamples for training, analyzing the role of the filtering mechanism in controlling pseudosample quality. The third group, *w/o stage-3 generation* (w/o SG), uses only the pseudosamples generated in the first stage for training, omitting the third-stage regeneration process to verify the value of iterative optimization in improving semantic consistency and distribution fitting of pseudosamples. The fourth group, *direct mix after filtering* (DMF), directly mixes the retained pseudosamples with the source domain samples for training after first-stage generation and second-stage similarity filtering (with a fixed threshold of 0.5), skipping the third-stage diffusion generation, to explore whether further regeneration optimization after filtering is necessary.

The results are shown in Tables VII and VIII. Overall, the complete model achieves the best performance across all tasks. Compared with the full model, removing the adversarial training module (w/o AG) results in a significant performance drop, demonstrating the critical role of the adversarial mechanism in the generation of target domain features. Removing the similarity filtering mechanism (w/o SF) also causes a notable decrease in performance, highlighting the importance of this module in controlling pseudosample quality and suppressing noise interference. When only one diffusion generation is performed without the third-stage regeneration (w/o SG), performance declines in most tasks, showing the value of the secondary generation in further optimizing semantic consistency and distribution alignment. In the direct mix after filtering strategy (DMF), although some tasks show good performance, overall it is slightly inferior to the complete model, indicating that a single generation and filtering step is insufficient for high-quality cross-domain transfer.

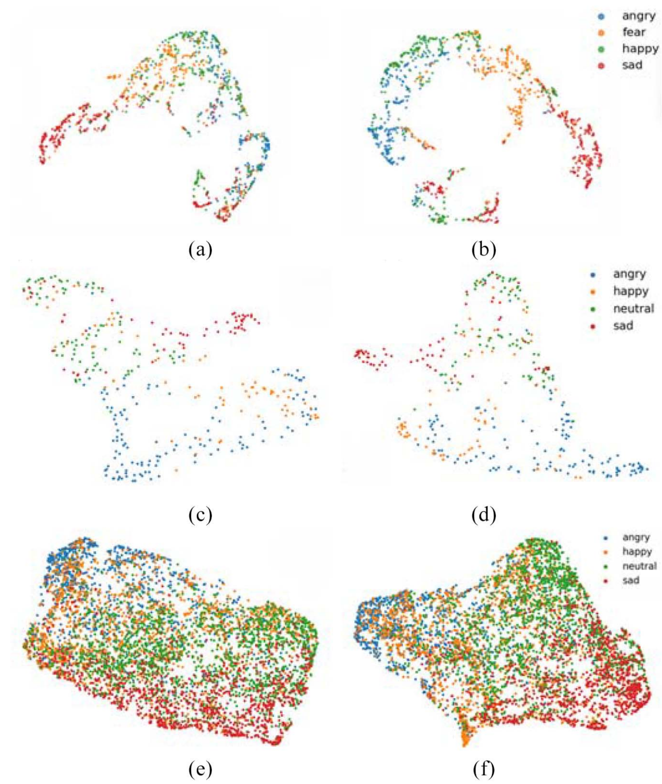

Fig. 8. UMAP visualization of feature distributions. (a) Baseline method on C-B task. (b) DFMDM on C-B task. (c) Baseline method on I-B task. (d) DFMDM on I-B task. (e) Baseline method on B-I task. (f) DFMDM on B-I task.

E. Feature Visualization

To provide a more intuitive comparison of the effectiveness of the proposed method, we visualize the feature distributions of the final UDA models using uniform manifold approximation and projection (UMAP). Specifically, the feature representations are extracted from the output of the last hidden layer of the trained UDA models. During the UMAP dimensionality reduction process, the Euclidean distance (i.e., L_2 distance) is adopted to measure the similarity between feature vectors, which is used to characterize their relative relationships in the embedding space. To ensure fairness and comparability, all methods employ the same distance metric and identical UMAP parameter settings.

We provide visualization results under multiple cross-corpus settings, including C→B, I→B, and B→I. Taking the C→B experimental setting as an example, the visualization results are illustrated in Fig. 8. Compared with the original MRAN, the proposed DFMDM significantly enhances the discriminative capability on the target domain by introducing high-quality pseudosamples into the source domain.

From the class-wise distributions, the proposed method exhibits clear and compact clustering patterns for the *sad*, *angry*, and *fear* categories, indicating improved feature separability for these emotions. In contrast, the *happy* category presents a relatively scattered distribution, reflecting weaker clustering behavior, which is consistent with the corresponding confusion matrix results.

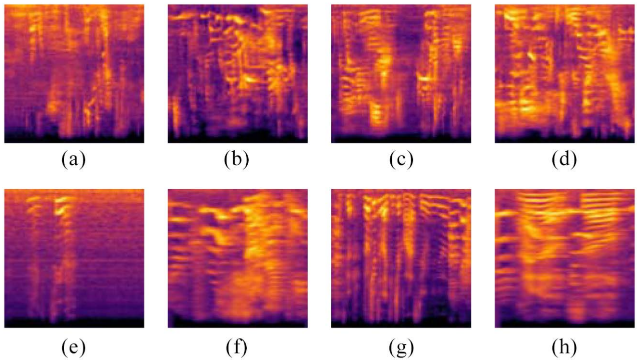

Fig. 9. Real samples and generated pseudosamples are displayed in lower and upper row, respectively. (a) Sad; (b) angry; (c) neutral; (d) happy; (e) sad; (f) angry; (g) neutral; and (h) happy.

In addition, we further provide a visual comparison between real samples and generated pseudosamples, as shown in Fig. 9. The pseudosamples generated by the proposed method exhibit similar distribution characteristics to real target-domain samples, which further validates the effectiveness of the proposed domain-guided diffusion strategy in generating discriminative and domain-consistent pseudodata.

V. Conclusion

To address the issue of ineffective control over pseudosample quality in existing cross-database SER methods, this article proposes a DFMDM. Experimental results demonstrate that the proposed method significantly outperforms mainstream existing approaches, particularly exhibiting stronger robustness and generalization ability in tasks with imbalanced target domain sample distributions.

Although this work achieves notable progress in cross-database SER, certain limitations of DFMDM should be acknowledged. First, the multistage diffusion generation process incurs substantial computational overhead, which may hinder practical deployment. Second, the model is sensitive to high-quality spectrogram features, and variations in feature preprocessing across databases may affect robustness. Future research may explore more lightweight diffusion architectures or end-to-end representation learning approaches to curtail computational expenditures while upholding performance standards. Moreover, the current method primarily relies on unimodal information; future work may consider integrating multimodal emotional features to enhance the comprehensiveness and robustness of emotion recognition.

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Jingjie Yan received the B.E. degree in electronic science and technology and the M.S. degree in signal and information processing from China University of Mining and Technology, Beijing, China, in 2006 and 2009, respectively, and the Ph.D. degree in single and information processing from Southeast University, Nanjing, China, in 2014.

He is currently an Associate Professor with the College of Communication and Information Engineering, Nanjing University of Posts and Telecommunications (NUPT), Nanjing. His research interests include pattern recognition, affective computing, computer vision, and machine learning.

BoYan Sun received the B.Eng. degree in electronic and information engineering in 2024 from the College of Communication and Information Engineering, Nanjing University of Posts and Telecommunications, Nanjing, China, where he is currently working toward the Ph.D. degree in information and communication engineering.

His research interests include speech emotion recognition and affective computing.

YueBo Yue received the B.Eng. degree in electronic science and technology from the School of Electrical Engineering and Automation, Luoyang Institute of Science and Technology, Luoyang, China, in 2022, and the M.S. degree in signal and information processing from the College of Communication and Information Engineering, Nanjing University of Posts and Telecommunications, Nanjing, China, in 2025.

His research interests include transfer learning and computer vision.

Xiaoyang Zhou received the B.E. degree in information engineering and the M.S. degree in signal and information processing from Southeast University, Nanjing, China, in 2009 and 2012, respectively.

He is currently the Chief Technology Officer (CTO) with China Mobile Zijin (Jiangsu) Innovation Research Institute Company, Ltd., Jiangsu, China. His research interests include pattern recognition, trusted computing, and federated learning.

Ying Liu received the B.E. degree in electromagnetic field and wireless technology and the M.S. degree in signal and information processing from Nanjing University of Posts and Telecommunications, Nanjing, China, in 2010 and 2013, respectively.

She is currently working with China Mobile Communications Group Jiangsu Company, Ltd., Nanjing. Her research interests include deep learning, cybersecurity, data security, and information security.